

Anomaly Detection in Countermovement Jump Data

Dartmouth Athletics
Travis Owusu | May 2026

Overview

VALD ForceDecks dual force plate technology captures countermovement jump (CMJ) biomechanics for athletes across 23 sports at Dartmouth Athletics. These tests are used by strength and conditioning staff to monitor athlete readiness, track training load responses, and identify performance trends over the course of a season.

During testing, athletes select their name from a dropdown menu before jumping. On occasion, the wrong name is selected, causing one athlete's jump data to be recorded under a different athlete's profile. This data misattribution problem silently corrupts trend analysis, making it impossible to reliably track individual athlete performance over time.

This project built a machine learning pipeline to automatically detect these mislabeled tests across more than 23,000 records spanning January 2024 through March 2026. The goal was to flag tests that are inconsistent with an athlete's established profile so the strength and conditioning staff can verify and correct them.

Data

Data folder:

- `forcedecks_anomaly_data_full`: 23,405 CMJ test records across 835 unique athletes and 23 sports, with 25 biomechanical metrics per test including body weight, jump height, peak power, force at zero velocity, eccentric duration, and bilateral asymmetry measurements.
- `forcedecks_anomaly_data_simplified`: 18,059 records across 583 athletes containing the core metrics used for modeling: body weight, jump height, and force at zero velocity.
- `Forceplate_Invalid tests & Forceplate_Invalid_Tests_Olympic`: Two Excel files manually compiled by Dartmouth strength and conditioning staff identifying 131 confirmed mislabeled tests across all sports. These served as the ground truth labels for model training and evaluation.

The labeled anomaly files were provided by the S&C staff who identified erroneous tests through observation during testing sessions. These labels represent confirmed anomalies rather than a complete census, additional mislabeled tests likely exist that were never caught manually. This is the problem the model was designed to address.

Cleaned_data folder:

- d3.csv: The original general label file containing 131 confirmed anomalies across 111 unique athletes, spanning January 2024 through March 2026. Each row identifies a specific test by athlete name, date, and set number that was manually flagged by the strength and conditioning staff as mislabeled. This file served as the primary source of ground truth labels for model training and evaluation.
- d3_updated.csv: An expanded version of d3 that includes 148 confirmed and model-verified anomalies across 121 unique athletes. The additional 17 entries beyond the original 131 represent high confidence model predictions that were validated against physiological plausibility criteria, specifically body weight changes that exceeded realistic thresholds given the time elapsed since the athlete's last session. This file was used as the evolving label set across iterative retraining rounds.
- d4.csv: The Olympic athlete label file containing 76 confirmed anomalies across 68 unique athletes. This file includes a sport column with full sport names and was used alongside d3 during model development. All 68 athletes in d4 also appear in d3, meaning d4 represents a subset of the broader confirmed anomaly pool rather than an independent set.
- fulld1.csv: The complete cleaned ForceDecks dataset containing 23,405 test records across 835 unique athletes from January 2024 through March 2026. Each row represents a single jump rep and includes body weight, jump height, force at zero velocity, set number, days since last test, and all engineered change features. The is_anomaly column is set to 0 across all rows in this file, it represents the full dataset prior to label assignment and serves as the base from which training, testing, and prediction sets were constructed.
- full_prediction_dataset: The final model output containing all 23,405 test records with predictions from both the Hist Gradient Boosting and Isolation Forest models. Each row includes the original test metrics, model confidence scores, individual model predictions, and a model_indicated_anomaly column that flags the 71 tests where both models agreed a record was anomalous. This is the primary deliverable for the strength and conditioning staff, a complete view of the dataset with all model-predicted anomalies surfaced for verification.

Approach

Feature Engineering

Raw test metrics alone are insufficient to identify anomalies. A body weight of 180 lbs is normal for one athlete and suspicious for another. The key insight is that anomalies are best detected through change over time rather than absolute values. A 40 lb weight drop over two days is physically impossible. The same drop over four months may be legitimate.

The following features were engineered for each test record:

- Body weight change: difference in body weight from the athlete's previous session.
- Body weight change per day: rate of body weight change normalized by days elapsed since last session.
- Jump height change: difference in jump height from the athlete's previous session.
- Jump height change per day: rate of jump height change normalized by days elapsed.
- Days since last test: number of days between the current session and the athlete's previous session.

Train and Test Strategy

A key challenge in this project was preventing data leakage, the situation where the model is evaluated on data it has already learned from, producing artificially strong results. To address this, athletes were divided into completely separate groups for training and testing. No athlete appeared in both sets.

Athletes with at least one confirmed anomaly label were used for modeling. An 80/20 split was applied at the athlete level, with 88 athletes used for training and 23 for testing. The training set included both confirmed anomaly rows and unlabeled rows for each athlete, giving the model examples of both what an error looks like and what a normal test looks like for a given individual (Knowing that some of the unlabeled tests are still anomalies).

After training and testing, the model was applied to the remaining athletes in the full dataset, those with no confirmed labels, to generate new anomaly predictions for review.

Models

Three models supervised were trained and compared:

- Logistic Regression: a simple linear classification model used as a baseline. Produced too many false alarms to be practical.
- Random Forest: an ensemble of decision trees. Precise but conservative, missing too many real anomalies.
- Hist Gradient Boosting (HGB): the primary model. Performed best overall with the strongest balance between catching anomalies and minimizing false alarms.

An Isolation Forest, an unsupervised anomaly detection algorithm that requires no labels, was used as a fourth model. Because it learns from the data structure rather than from labels, it serves as an independent check. When HGB and Isolation Forest both flag the same test, the combined signal is more credible than either model flagging it alone.

After model fitting, feature influence was assessed using feature importance, absolute coefficient importance, and permutation importance analyses. Results consistently showed that bodyweight change and jump height change were among the highest-ranking predictors of anomalous observations across models.

Iterative Retraining

A big issue was the amount of anomalies that were given to work with. To solve this issue, the model was designed as an iterative feedback loop rather than a one-time analysis. After each round of predictions, high confidence flags were reviewed against physiological plausibility criteria (mainly bodyweight with respect to). were added back to the label set and the model was retrained on the expanded data. This process was repeated across 2 rounds, progressively improving the model's ability to detect anomalies, but stopped after 2 rounds to not be over ambitious.

Results

Model Performance

The table below shows final performance of the Hist Gradient Boosting model on the held-out test set after iterative retraining:

Metric	Clean Tests	Anomalies
Precision	0.99	0.43
Recall	0.97	0.76
F1 Score	0.98	0.55
Support	1,248	34
Accuracy	.97	

Precision of 0.43 on anomalies means roughly 1 in 2 flags raised by the model is a genuine anomaly. Recall of 0.76 means the model successfully identified approximately 3 out of every 4 real anomalies in the test set missing 8 values. For a dataset where the true anomaly rate is below 1%, these results represent a meaningful improvement over manual review.

Anomaly Predictions

After iterative retraining, the final model was applied to the full dataset. Predictions were tiered into three confidence levels based on model agreement:

- High confidence: all three models agree the test is anomalous. These cases have the strongest physiological evidence and should be prioritized for staff verification.
- Medium confidence: at least HGB and Isolation Forest agree. These are credible but warrant a closer look before being confirmed.
- HGB only: flagged by the primary model alone. Treat as weak signals requiring staff judgment.

Notable findings from the prediction phase included Chase Engdahl, whose profile was flagged across nine separate sessions spanning October 2025 through February 2026 with consistent body weight drops of 17 to 60 lbs, suggesting a systematic recurring misattribution rather than a one-off error. Carson Franks was flagged with a body weight change of -198.5 lbs in two days, representing a near-certain anomaly.

Limitations

- Label completeness: the 131 confirmed anomalies provided by the S&C staff represent observed errors, not a complete census. Some unlabeled tests in the training data may be genuine anomalies, which means the model was inadvertently trained on some incorrect examples. This is an inherent limitation of the labeling process and is why the iterative verification loop is important.
- Small labeled dataset: with only 33 anomalies in the test set, performance metrics have meaningful variance. Small changes in predictions on a handful of rows can shift recall and precision noticeably.
- Same-weight misattributions: the model relies heavily on body weight deviation. Cases where the wrong athlete has a similar body weight to the correct athlete may go undetected regardless of model sophistication.
- Positive-unlabeled learning: because only confirmed positives exist in the label file and all other tests are assumed clean, the model cannot distinguish between a truly clean test and an undiscovered anomaly in the unlabeled data.

Next Steps

- S&C staff verification: review the high confidence flags generated by the model and confirm or dismiss each one. Confirmed flags should be added to the label set to improve the next training round.
- Continued iterative retraining: each round of verification improves the model. The feedback loop is already in place and can be run with minimal effort as new verifications come in.
- Asymmetry feature integration: the full ForceDecks dataset includes five bilateral asymmetry metrics that were not used in this model. These could help detect anomalies where body weight is similar between athletes but movement patterns differ, addressing the same-weight limitation described above.
- Expansion to all athletes: the current model was trained and tested on athletes with confirmed labels. Applying it systematically to all 835 athletes in the full dataset will surface the broadest possible set of potential anomalies for staff review.
- Clean dataset delivery: once verified anomalies are removed from the dataset, the S&C staff will have a reliable foundation for the seasonal trend analysis and cross-season performance comparisons described in the original project brief.
- A physiological plausibility filter can also be applied to new model predictions, cross-referencing body weight change against days elapsed to flag only changes that exceed realistic thresholds for a given time window.